PRACTICE ASSIGNMENT
OBJECT ORIENTED PROGRAMMING

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Course : Object oriented programming with C++
Group : 1MCA3

1. **Write a program to add two objects of different classes using a friend function.**

```
#include <iostream>
using namespace std;

class B;
class A {
   int x;
public:
   void getA() {
      x = 10;
   }
   friend int add(A, B);
};
class B {
   int y;
public:
   void getB() {
      y = 20;
   }
   friend int add(A, B);
};
int add(A a, B b) {
   return a.x + b.y;
}
int main() {
   A obj1;
   B obj2;
   obj1.getA();
   obj2.getB();
   cout<<"Sum = "<<add(obj1, obj2);
   return 0;
}
```

```
Sum = 30
--------------------------------
Process exited after 0.1224 seconds with return value 0
Press any key to continue . . .
```

2. **Create a friend class that can access private members of another class and display them.**

```
#include <iostream>
using namespace std;
class A {
```

```
    int x;
public:
    A()
    {
        x = 100;
    }
    friend class B;
};
class B {
public:
    void display(A obj)
    {
        cout << "Value of x = " << obj.x;
    }
};
int main() {
    A obj1;
    B obj2;
    obj2.display(obj1);
    return 0;
}
```

```
Value of x = 100
--------------------------------
Process exited after 0.2574 seconds with return value 0
Press any key to continue . . .
```

3. **Use a pointer to a class object to call its member functions.**

```
#include <iostream>
using namespace std;
class Student {
public:
    void display()
    {
        cout << "Hello Student";
    }
};
int main() {
    Student obj;
    Student *ptr;
    ptr = &obj;
    ptr->display();
    return 0;
}
```

```
Hello Student
--------------------------------
Process exited after 0.2478 seconds with return value 0
Press any key to continue . . .
```

4. **Write a program that uses the this pointer to print the address of the current object.**

```
#include <iostream>
using namespace std;
class Demo {
public:
    void display()    {
        cout << "Address of current object = " << this;
    }
};
int main() {
    Demo obj;
    obj.display();
    return 0;
}
```

```
Address of current object = 0x70fe4f
--------------------------------
Process exited after 0.2649 seconds with return value 0
Press any key to continue . . .
```

5. **Allocate an array of objects dynamically and delete it.**

```
#include <iostream>
using namespace std;
class Student {
public:
    int roll;
    void getData()    {
        cin >> roll;
    }
    void display()    {
        cout << "Roll number = " << roll << endl;
    }
};
int main() {
    Student *ptr;
    ptr = new Student[3];
    cout << "Enter 3 roll numbers: ";
    for (int i = 0; i < 3; i++)    {
        ptr[i].getData();
    }
    cout << "\nStudent Details:\n";
    for (int i = 0; i < 3; i++)    {
        ptr[i].display();
    }
    delete[] ptr;
    return 0;
}
```

```
Enter 3 roll numbers: 11 12 13

Student Details:
Roll number = 11
Roll number = 12
Roll number = 13

--------------------------------
Process exited after 24.36 seconds with return value 0
Press any key to continue . . .
```